

Vrinda Venugopal M

B.Tech Computer Science Student

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PROFILE

Computer Science undergraduate with a strong foundation in software engineering, systems, and artificial intelligence. Experienced in building technically diverse projects and conducting research on LLM evaluation and model selection. Driven by solving complex engineering problems and building efficient, scalable systems.

EDUCATION

B.Tech in Computer Science and Engineering , PES University CGPA: 8.45	2024 – 2028
Class XII , BASE PU College 95.5%	2024
Class X , Mitra Academy 98.2%	2022

TECHNICAL SKILLS

Languages

C, C++, Python, JavaScript, Dart (Learning)

Data & Databases

SQLite, MongoDB, NumPy, Pandas

Frameworks and Technologies

FastAPI, LangGraph, React.js, Node.js, Express.js, CodeMirror, PyTorch, Flutter (Learning)

Systems and Networking

Linux, TCP/IP, OpenFlow, Software-Defined Networking, Mininet, Ryu, WebSockets

AI and Machine Learning

Agentic AI, LLM Agents, ReAct, Tool Calling, Sentence Transformers, Random Forest

Tools & Platforms

Git, GitHub, Grafana, VS Code

WORK EXPERIENCE

Research Intern - Centre for Computer Networks and Cyber Security (CCNCS), PES University	June 2026 – July 2026
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PROJECTS

CaskDB

- Engineered a persistent key-value storage engine with write-ahead logging, crash recovery, sorted memtables, immutable SSTables, and multi-file compaction for durable disk-backed storage.
- Tech: C++17, Write-Ahead Logging (WAL), SSTables, File I/O | GitHub: <https://github.com/Vrinda0211/CaskDB>

Vertex

- Built a real-time collaborative code editor with concurrent state synchronization, Operational Transformation for conflicting edits, and a context-aware AI pair programmer with streamed responses.
- Tech: Python, FastAPI, WebSockets, Operational Transformation, CodeMirror, Gemini API | GitHub: <https://github.com/Vrinda0211/Vertex>

Multi-Agent SOC Analyst Assistant

- Developed an agentic cybersecurity orchestration system coordinating specialized Triage, Investigation, and Response agents with stateful workflows, dynamic tool selection, and security intelligence enrichment.
- Tech: Python, LangGraph, ReAct, FastAPI, SQLite, Grafana, Gemini API, MITRE ATT&CK | GitHub: <https://github.com/Vrinda0211/AgenticAI-MultiAgent-SOC>

CodeDistill

- Built a Chrome extension that extracts executable code from noisy web content using deterministic text normalization and AI-assisted code extraction for copy-ready source code.
- Tech: JavaScript, Chrome Extension APIs, Prompt Engineering | GitHub: <https://github.com/Vrinda0211/CodeDistill>

SDN Link Failure Recovery

- Developed a fault-tolerant SDN controller that detects link failures, recomputes shortest paths, invalidates stale flow rules, and dynamically reroutes traffic through alternate network paths.
- Tech: Python, Ryu, Mininet, OpenFlow, BFS, Software-Defined Networking | GitHub: <https://github.com/Vrinda0211/SDN-Link-Failure-Recovery>

Lightweight Container Runtime

- Built a minimal Linux container runtime with process lifecycle management and kernel-level resource monitoring.
- Tech: C, Linux Kernel Modules, System Programming | GitHub: <https://github.com/Vrinda0211/Container-Runtime>

PUBLICATIONS

A Difficulty-Aware Prompt-Model Recommendation Framework for LLM-Generated Python Code Using Competency-Based Evaluation, IEEE CONECCT Conference (accepted for publication)